

Detection of Fake Products using Blockchain

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Abstract— Manufacturing and selling of counterfeit items and goods endangers end consumers' financial, health, and safety. Additionally, it affects the economic expansion of original manufacturers and businesses through revenue loss, product slander, downtime, replacement costs, the need for brands to spend money fending off counterfeits, sales theft, etc. To combat these critical counterfeiting consequences, a blockchain-based system is utilized to identify original items and detect forged products to secure the identification of genuine commodities. Any application that uses Blockchain technology as its foundation architecture assures that its data is tamper-proof. The product's QR code is linked to a block chain in order to preserve product information and guaranteed unique codes as blocks in the database, and fake items are detected using a camera scanner. If the codes match, the consumer will receive a notice attesting to the authenticity of the items; otherwise, they will receive a notice informing them that the product is fake. This technique assures that consumers will not rely solely on retailers to identify whether items are genuine or counterfeit.

Keywords— *Counterfeit detection, Blockchain technology, Product authenticity, QR code*

I. INTRODUCTION

In the dynamic landscape of global technological and product development, businesses face significant risks, particularly in dealing with issues such as counterfeiting and forgery. These challenges pose threats to a company's reputation, sales, and overall customer satisfaction [1]. Within the complex web of the supply chain, distinguishing between authentic and counterfeit goods becomes a critical task for manufacturers, who often grapple with substantial losses due to the proliferation of fake products.

Addressing this issue, blockchain technology emerges as a powerful tool to ensure the authenticity of products. Blockchain, essentially an information protection mechanism, introduces a high level of resistance against unauthorized updates, hacks, or manipulations of the system. Operating as a decentralized and replicated digital ledger of

transactions, each block in the blockchain encompasses multiple transactions. When a new transaction occurs, it is added to the records of all participants in the network [2]. This technology falls under the broader category of Distributed Ledger Technology (DLT), characterizing a decentralized database maintained by a large network of participants [3]. Utilizing an immutable cryptographic signature, known as a hash, blockchain records transactions securely.

Crucially, blockchain extends beyond its association with cryptocurrency and holds diverse applications in various sectors, including government, finance, banking, accounting, and business processing. By leveraging blockchain technology, users and customers eliminate the need for third-party verification of product security [5]. One notable application of blockchain is its efficacy in combating the proliferation of counterfeit products, offering enhanced security measures.

The security inherent in blockchain technology is exemplified by the creation and utilization of a unique hash code for each product stored on the network. This hash code, generated for every product transaction, enables the storage of a comprehensive record of transactions for each product and its current owner. In the proposed system, each product is assigned a QR code, allowing end-users to scan and access detailed product information. This innovative approach not only enhances security but also provides a practical means for consumers to verify the authenticity of products by simply scanning the QR code [6]. In essence, blockchain technology proves to be a robust solution to the persistent challenge of counterfeit products, offering heightened security measures and a transparent means of product authentication.

II. LITERATURE SURVEY

The literature surveyed on product authentication and anti-counterfeiting strategies highlights a prevalent theme of leveraging advanced technologies, particularly blockchain, to ensure the integrity and traceability of products in supply chains.

One common approach involves assigning public and private keys as QR codes to products, with scanning apps possessing cryptographic functionality for decryption. Manufacturers play a crucial role by maintaining servers that validate buyer requests and match item codes. This system enhances traceability through a Cyber-Physical System (CPS)-based supply chain management architecture, promoting supply chain monitoring and data analytics for improved product management [7].

Another explored solution focuses on brand protection and anti-counterfeiting using smart tags and Cloud-enabled technologies. This entails utilizing quick response codes and functional inks, facilitating two-way communication between manufacturers and end-users [8]. Blockchain technology emerges as a decentralized solution to reduce reliance on merchants for product authentication [9]. A study introduces a Blockchain-Based Management (BCBM) system, linking product barcodes to the blockchain for counterfeit identification using barcode readers. This system stores product information as database blocks, verifying customer codes against blockchain entries [10].

Recognizing the urgency in addressing the rise of counterfeit items in online and black marketplaces, the literature explores various methods for detection. Blockchain is proposed to enhance supply chain management, presenting a paradigm shift in blockchain supply chain management to improve acquisition accuracy [11].

A comprehensive framework for information resource management in distributed organizations is proposed, emphasizing distributed management theory, corporate management, and spatial management. The supply chain, viewed as a linked network from raw material distribution to end consumer delivery, employs an anti-product forgery system using Ethereum-based Blockchain architecture. The transparency features of blockchain ensure that end consumers can independently verify the provenance of purchased items without relying solely on third-party verification [12]. Overall, the literature underscores the critical role of technological innovations, particularly blockchain, in fortifying supply chains against counterfeit threats and enhancing transparency for end consumers.

III. METHODOLOGY

This project involves distinct user scenarios for manufacturers, sellers, and customers, each accessing customized login pages. Manufacturers log in to input product details, and the application automatically generates a unique QR code for every added product stored in the database. These products are then transacted to sellers, acting as intermediaries between manufacturers and end customers. Sellers, upon login, can sell specific products to customers and showcase their available product list, each item equipped with a QR code sourced from the database.

For end users or customers, after purchasing a product from the seller, they can authenticate its genuineness. The customer can utilize the QR code to validate the product's authenticity and also access their purchase history. The functionalities within this system can be categorized into the following modules as in figure.1.

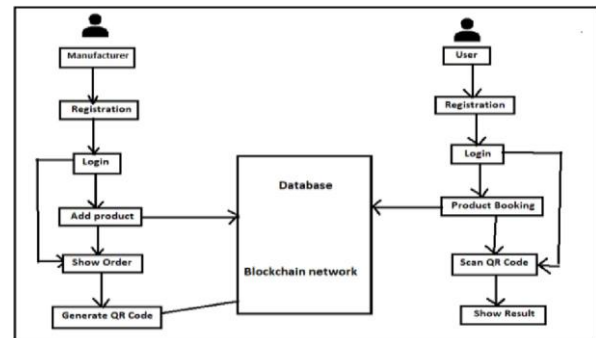


Fig.1. System Design/Workflow

- **Sign Up/Login:** Users are required to sign up and log in to the website, accessing their role-specific features.
- **Add Product:** Manufacturers, upon logging in, can seamlessly add products utilizing JavaScript and smart contracts within the blockchain database.
- **Generate QR Code:** Product data is displayed in the admin section, where administrators verify and generate QR codes for the corresponding products using JavaScript, subsequently adding them to the blockchain database.
- **Scan QR Code:** Customers can utilize an Android application to scan the QR code, triggering the display of comprehensive product information on their mobile screens.

IV. IMPLEMENTATION

This study has employed Metamask for wallet creation and transaction management, VS Code as the platform to execute our programs, Truffle as the testing framework for blockchain applications, and Node.js for project development. The initial phase involved constructing the application's frontend, focusing on the manufacturing section. Ganache, serving as the blockchain network, simulated the Ethereum blockchain for smart contract interactions and facilitated the creation of a local blockchain.

To interface with the application, an account was established in Metamask. In VS Code, Ganache was utilized to create a local blockchain. Configuring Ganache involved setting up a development mode network on port 7545, enabling client connections. Initiating the Ganache server resulted in the creation of ten accounts, each assigned individual private keys and 1000ETH.

Following this, the command prompt was used to execute the following commands:

- Smart Contract Compilation: To compile smart contracts, the truffle compile command was executed.
- Smart Contract Migration: The next step involved migrating smart contracts to the Ganache network, achieved through the truffle migrate command. Upon completion, the block count increased from 0 to 4.

This process facilitated the establishment of a functional local blockchain using Ganache, enabling seamless interaction with our application.

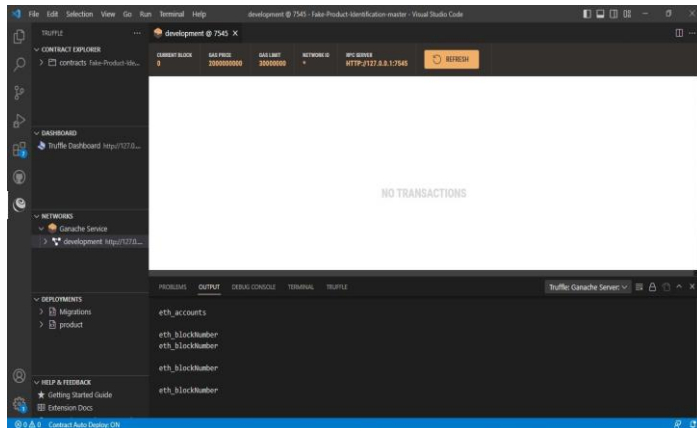


Fig.2: Initial Count of block in Ganache

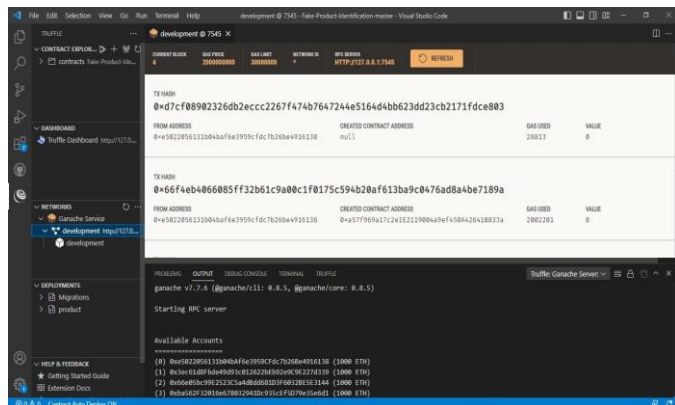


Fig.3: Updated Count of block in Ganache

Following that, we must login to the metamask to build a local host using parameters such as network id, RPC server, and chain code. A default account named Account1 has been created for the local host, and another account named Account2 is then imported using the private key displayed in the vs code output.

The image below depicts the metamask account, where the ETH linked with that account is presently 0 because the value will alter whenever the program executes. Multiple accounts can be made for a metamask, with no limitation on the amount of accounts that one login can have.

The private key one per account will be concealed, while the public key would be visible to all. In our project, the private key will be known only to the individual who has the code in their possession along with the metamask login

details; if those two are not accessible, there won't be any means of obtaining the private key for accounts.

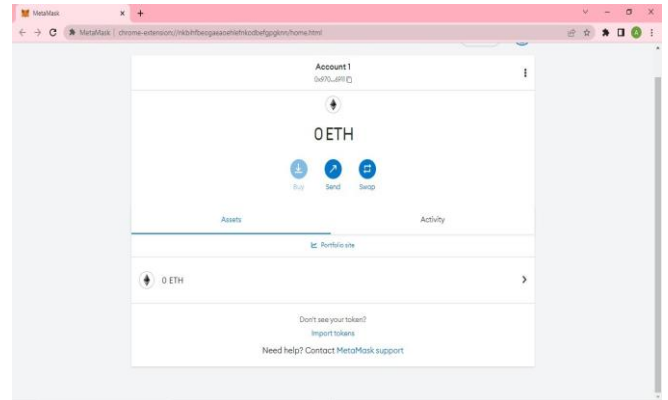


Fig.4: Homepage of Metamask

After importing Account2 holds around 999.9926 ETH (ETH is an acronym for Ethereum). We have to transfer an insignificant amount of ETH from account2 to account1 in order to link the metamask and the local blockchain on the ganache server. Subsequently to launch the project, use the command **npm run dev** in the command prompt, and the homepage will be opened in our default browser, which in our case is Google Chrome.

The below figure depicts the main screen of the fake product identification. The accessible choices are the home, the manufacturer, the seller, and the client. The manufacturer has additional subheadings that require a few details to be entered and filled out by the manufacturers end.

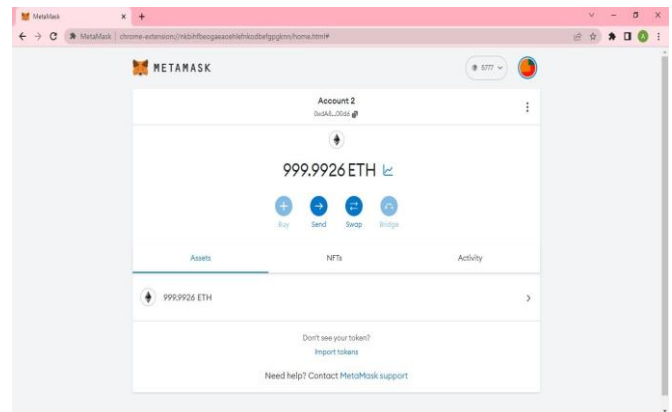


Fig.5: Imported account in Metamask

The manufacturers segment is where the product is created before being delivered to the seller. The sellers section is entirely dedicated to the seller who is selling that product to that specific customer, while the customer segment is dedicated to the consumers who are purchasing the product.

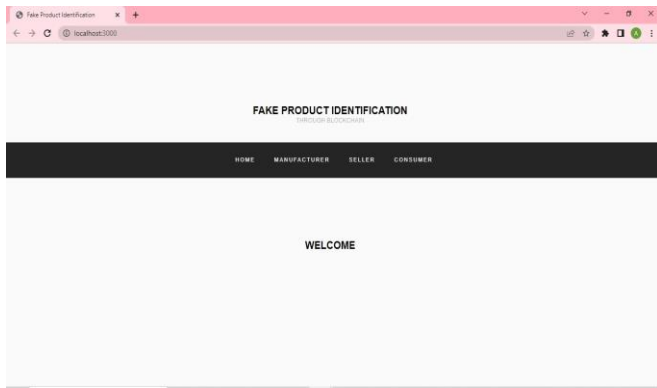


Fig.6: Homepage of the website

To link the webpage to the local blockchain, as illustrated in the figure below, we must connect the webpage to the metamask.

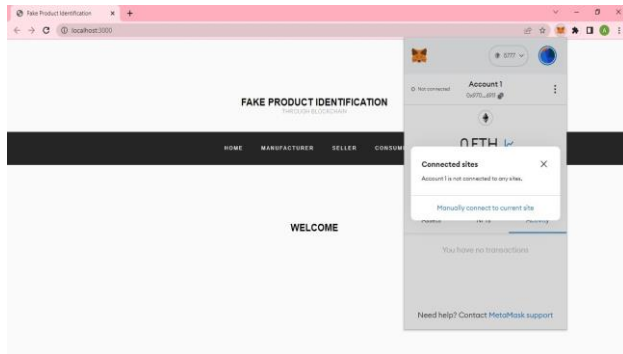


Fig.7: Connecting Metamask to the website

In the below image, the website is connected to the metamask. Here, the Account1 is connected to the <https://localhost:3000> where the localhost 3000 indicates the website is running on the computer which is hosting the application.

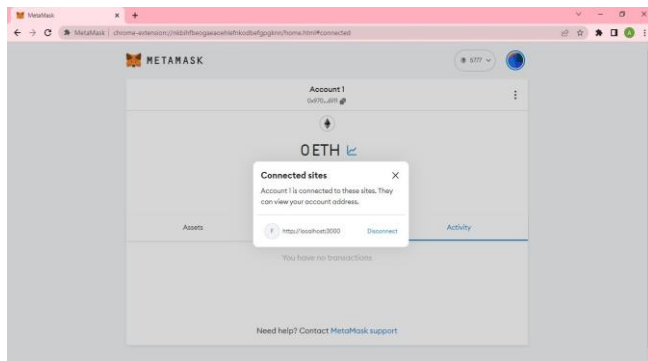


Fig.8: Metamask connected to website

Once the website is linked to Metamask, we're able to utilise the application to add sellers and items, as well as determine whether a product is fraudulent or not.

The first portion, which is the manufacturer area, is divided into four subsections: add product, add seller, sell product to seller, and query seller. The second portion is the seller section, which includes two subsections: sell goods to customer and items for sale. The last portion is a customer section that includes two subsections: consumer purchase history and product verification.

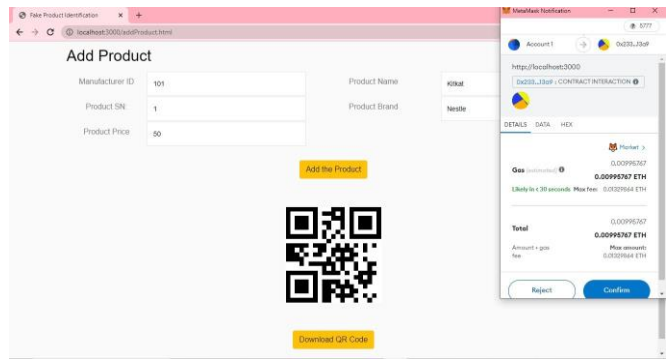


Fig.9: Adding product details

The following is an explanation of each subsection in the manufacturing section:

ADD PRODUCT: The manufacturer will enter information for the manufacturer ID, product SN, product price, name, and brand when we add a product. After entering all of these parameters, an automated QR code will be created that is unique to that product SN and brand. The blockchain will produce the QR code in the webpage. In the above image, the add product subsection is shown along with the QR code that is generated.

- **ADD SELLER:** The information for the manufacturer ID, seller name, code, manager, brand, phone number, and address in add seller is entered. The producer can add many vendors for one product because the sellers will differ depending on region and other factors.
- **SELL PRODUCT TO SELLER:** The product will be scanned and the seller code is entered.
- **QUERY SELLER:** This feature is used to get the data of how many sellers are associated with a particular manufacturer.

The following is an explanation of each subsection in the seller section:

SELL PRODUCT TO CONSUMER: The product will be scanned and the consumer code is entered.

- **PRODUCTS FOR SALE:** This feature is used to get the details of the product associated with a particular

The following is an explanation of each subsection in the consumer section:

CONSUMER PURCHASE HISTORY: It will provide the details of the products purchased by the consumer.

- **PRODUCT VERIFICATION:** This will determine if the product sold to the consumer is fraudulent or genuine, the result will be shown once the QR code is either scanned with the help of a camera or by uploading the image of the QR code that was downloaded when it was generated by the blockchain.

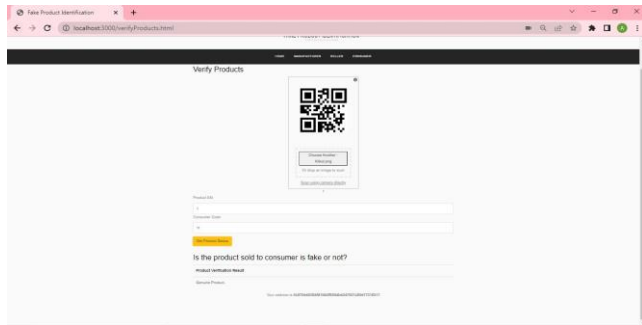


Fig.10: Product verified as Genuine



Fig.11: Product verified as fake

CONCLUSION

This study proposed a robust anti-product forgery system based on blockchain technology, offering users a reliable solution to mitigate concerns about counterfeit goods by incorporating a minimal transaction charge. The encoding of product information into QR codes facilitates easy identification of fake goods through scanning, ensuring that digital information about items is securely stored in the form of blockchain blocks. Customers can thus have confidence in the quality of their purchases, leading to a reduction in the forging of branded goods and providing enterprises with a streamlined method to instill consumer confidence in the authenticity of their products.

The proposed approach not only contributes to the fight against counterfeit items but also fosters trust and goodwill

between manufacturers and consumers. This trust-building mechanism has broader implications for economic growth and the reduction of corruption.

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